enjoy finding ways in which two plus two is not equal to four. Much of life is not additive. If two mouthfuls of a food are pleasant, it does not follow that four mouthfuls will give twice as much pleasure. The pleasure may decrease even more with six mouthfuls. Twelve mouthfuls might be unpleasant. "I remember once discovering a life situation in which two plus two equaled zero," Scott says.

The free minds of the twenty-first century challenge everything that seems self-evident. They like to try on mentally different points of view. They search for their hidden assumptions and delight in bringing them to the surface. They are experts at changing their minds. "There are many people I especially like because they do not share my points of view," Hella says. "I enjoy talking with them when they vigorously defend a position that contradicts mine. I know I learn more when I find people with ideas that challenge mine."

The Morning: Conference

Hella enjoys studying human relationships. She has asked the Correlation Center to indicate useful things she can do. She has been appointed to a committee that is studying the degree to which privacy in living areas adds or detracts from the human happiness potential. They are gathering data on what proportions of the population seem to achieve maximum fulfillment in those apartments that offer privacy versus those apartments that are shared by varying numbers of people. There is some evidence that self-selected groups of six people offer greater conversational variety, increased vicarious appreciations, and significantly enriched intellectual, esthetic, and sensory experiences. Larger groups may have a degree of superficiality and confusion that has drawbacks.

It is recognized that if each individual is to have a maximum of "Life, Liberty, and the Pursuit of Happiness," there must be opportunities from full privacy to full community participation in all activities. One should be able to choose whatever meets his needs best at the time. All planning is determined by the varied and changing preferences of individuals, not by what someone else thinks is "good for them."

Both Scott and Hella plan to confer with their respective colleagues on these scientific matters during the latter part of the morning. At the prearranged time the telescreens come to life. Scott in his portion of the room and Hella in hers talk with their associates throughout the world. A table is in the foreground of the screen and the people appear in the far-flung conference telescreens as though seated in a semicircle.

At one point in Scott's conference it is necessary to telecommunicate with a scientist on a space station. Later, they need the results of an experiment on hearing conducted eighty years previously. A request is made of the Correlation Center for this information. Within moments the data is presented upon the screen. The Correlation Center has recorded every book, document, and report that has ever been preserved in the history of the world. This information—classified, cross referenced, summarized, and evaluated—is available at all times. The proceedings of this conference are also recorded into the vast memory banks of the Correlation Center.

After their conferences Scott and Hella leave the main living area. Their chairs silently fold away and retract into the floor. They return to the balcony and recline upon a huge circular surface which automatically contours itself to their bodies and orients to the most suitable position to absorb filtered sunlight. The cybernator senses the mood of Scott and Hella and immediately provides a delightful fragrance of flowers with a background of stimulating music. Scott and Hella tenderly touch each other. He kisses her shoulders and she responds. They both begin to breathe deeply as they enjoy a multi-

dimensional experience that culminates in an ecstatic sexual climax.

As they relax automatic units gently massage their bodies. These units do not have any extensions or projections. They are electronic means that project gentle contractions into the musculature of the body. All points of the body can be gently or forcefully massaged at the same time, depending upon the training that one has given the cybernator that serves him. Scott and Hella's muscles are stimulated to develop and maintain strength. As the massage becomes gentler, it gradually lulls them to sleep.

Although they live in a voice-actuated environment designed to meet their needs, they are resourceful, capable individuals. They enjoy muscular activity. Since the cybernated machines protect them from boring, repetitive drudgery, they are unburdened and hence able to enjoy exertion. They often walk several miles in preference to using a cybernated car.

While Scott and Hella are asleep a friend calls. There is no jangling phone to awaken them, for the sensing devices in their environment know they are asleep and report this. The friend communicates his message to the three-dimensional recorder, which is promptly played for Scott and Hella when they awake.

Since it is possible to communicate quickly with any person on earth, no matter where he is, there is no problem in returning the call. They invited him to drop by that evening.

Scott and Hella feel a complete freedom to explore new areas of thinking and feeling. They want their values and interests to change so that they may experience a multi-dimensional life. As they seek newer and deeper areas of feeling and thinking, the cybernator senses these altered patterns and responds appropriately. Just as an English butler learned in previous centuries to pick up the moods and needs of his employers "intuitively," so the cybernators are designed to grasp totally the sensitivities of the people they serve. They sense almost in advance what each person will want. They furnish whatever concert, symphony, or other type of entertainment would best suit the need of each individual they serve. The cybernators never determine what humans should have or want. They always seek to provide what will best meet the needs of individuals from clues given by the pattern of past choices.

Although Scott and Hella are aware of the functions of the cybernator, they interact with it in a highly impersonal way. In the hidden mechanism in the walls, there are built-in, duplicate parts that automatically bypass any defective units in an instant without interruption of service.

The living patterns Scott and Hella prefer can be instantly transferred to any other living unit they visit. Their cybernator is connected to the Correlation Center so that any living unit throughout the world can immediately request instructions on the various patterns an individual prefers. Thus everyone in the twenty-first century feels at home wherever he is—in a space ship, an apartment in the Himalayas, or a living unit at the bottom of the sea. As Scott once expressed it, "The world is my home."

A Visit to Sumatra

During the afternoon Scott mentions a place in Sumatra he had once visited. On voice command, there

emanates from the walls a teleprojection of a scene from Sumatra. As these images change, Scott and Hella experience the smell of the forest, the green leaves of the jungle, and the animal life. These perfected teleprojection images can not be distinguished from the real objects.

The living area is at once transformed. Hanging vines and the lush tropical vegetation of the jungles of Sumatra appear in life-sized, three-dimensional color. Birds fly through their living area, and animals walk by, apparently within reach of Scott and Hella. They appear quite solid, and one can hear the flapping of bird's wings and the soft padding of an animal's paws. The sea life and beaches appear in full reality. With gentle breezes caressing their faces, Scott and Hella feel immersed in the breaking waves, and colorful tropical fish that surround them and occasionally seem to touch them. If there is anything they wish to see again or want enlarged, they have only to command the cybernator. They watch underwater plants shimmer past them and see reflections of sunbeams coming down through the water with iridescent shadows.

The newscast during the late afternoon briefly discusses signals received from outside the solar system. A particularly strong transmission is currently being monitored from a star some twenty-six light-years away, near the center of our galaxy. Apparently, intelligent beings are beaming strong signals toward earth. The linguistic system has not been cracked, but the computers are working on it. How soon will they contact other beings in the universe?

Sharing With Friends

Toward evening the cybernator alerts Scott that their friends will be there soon. Sonji and Jahn arrive in a pilotless levitator. It is propelled by highly energized ion particles that are emitted from the underside of the craft in a constant stream. This enables it to go up or down or be propelled in any direction desired. The levitator is not a flying machine. It does not depend upon air currents or any noisy unreliable devices such as propellers, wings, or ailerons. As it gently alights on the balcony, the guests appear to walk through its side, for there are no doors, cracks, or visible openings. Through verbal command the molecular bond of a portion of this outer skin yields to permit them to leave. The cybernator greets the guests and automatically guides them to the area where they are received by Scott and Hella. They make themselves comfortable on the responsive furniture.

These guests are not attracted to Scott and Hella for inconsequential chatter or the social consumption of alcohol. Jahn is engrossed in reconstructing biological organisms that roamed the earth in the distant eons of the past. Sonji is part of a team devising electronic means to propel rats, flies, and other undesired forms of life from a five-mile belt surrounding all of the cities of the globe. They excitedly exchange ideas pro, con, back, and forth.

Creative Recreation

There are many games with physical or intellectual challenge from which to choose. This evening Scott suggests they play a game of Intellectronics. The three-dimensional, teleprojection mechanism is turned on, and they eagerly don headsets with sensitive electronic pickups. The teleprojection area comes alive with a visual representation of the innermost feelings of each individual, somewhat similar to the

oscilloscope projection of vocal sounds. The screen is filled with a three-dimensional, infinitely varied spectrum of color. By interacting with each other, they make sprays of color that extend and dance in space before them. The forms blend with one another and merge as totally different and exciting patterns.

"What a novel thought! Who projected that? How surprising that this projection is so similar to mine." The game explores the innermost feelings of each participant. These electronic translations of the operation of the human mind are very meaningful to Scott, Hella, and their friends. If someone from previous centuries were to see them, they would appear abstract and meaningless. But to people who have had years of experience, this new imagery represents a sophisticated form of communication. Somewhat as an electroencephalogram operator in the twentieth century interpreted two-dimensional wavy lines, the participants of the new imagery are sharing a form of communication, involving the integration of motion and color interwoven in a symphonic sensorium.

Home Movies

After a pleasant evening with their friends, which has lasted until around four in the morning, Scott and Hella retire to their sleeping chamber. Suddenly, Scott remembers some three-dimensional color recordings he and Hella made in their sleeping chamber on a recent visit to the moon. Since the moon has only one-sixth of the gravitational pull of the earth, they weighed only about 20 pounds. With the full muscle strength designed for earth's gravitation, they were able to interact sexually with each other in ways that would be impossible on earth.

Upon command the cybernator projected these movies on the screen of their sleeping chamber while Scott and Hella enjoy again their previous delights. As the movie draws to a close, the cybernator uses its repertoire to enhance the sexuality of Scott and Hella. The whole spectrum of their sensations is keyed to this undisturbed act of love. The music keeps pace with their physiological activities and sensations. Temperatures are automatically maintained to meet their needs. All of their innermost feelings are accentuated and co-ordinated toward the gripping climax.

As they drift away into a restful sleep, the cybernator with its thousands of inputs throughout their entire apartment, maintains a constant surveillance over the well-being of Scott and Hella. They are the pioneers in a new age of social and individual symbiosis. It is an age in which a purposeful existence and a fulfilling life is shared equally by all. In six hours they will go to one of the fascinating underwater apartments built on a colorful Exuma reef. They sleep deeply to meet the opportunities of an exciting tomorrow.

9. A Multi-Dimensional Life

For some time Scott and Hella have been planning to visit an underwater resort in the Exuma Islands in the Bahamas. The cybernator automatically notifies the Correlation Center of their plans. Scott and Hella have no frenzy of last-minute packing to get ready for the trip. They simply leave with nothing in their hands. The entire world is their home, and they have no need to take anything with them. Food and drink will be available in the aircraft. Travellers in previous times were heavily burdened by baggage, but

anything Scott and Hella need will be available wherever they go. Their important papers, photographs and mementos are fed into the Correlation Center. They can be instantly retrieved any time or any place in the world. Soon retrieval on the moon will be possible.

On Hella's command the cybernator calls for a 1,000 mile-per-hour ion-propelled craft. Within minutes the pilotless craft stops on the landing area on top of their apartment building. They enter and by vocal command give coded directions for the apartments in the Exumas. As the craft races at supersonic speed over the highest clouds, Scott and Hella relax in contour chairs and watch the unfolding panorama of the cloud formations.

"Clouds are like hypotheses," muses Scott. "They're always changing."

"I find change exciting," Hella reflects. "I wouldn't want to live in a static society, where things are regarded as absolute and final."

"Well, I was thinking of the way men confused their notions of the world with reality," says Scott. "We know today that no theories are true or false—they are only more or less useful. They have more or less predictability."

"Didn't Einstein recognize this in his Theory of Relativity?" asks Hella.

"Exactly," says Scott. "Although it was fully accepted by scientists, he never said it was 'true.' He simply suggested that we use it if it has greater predictability than anything else. We'll throw it out if we can devise a theory that explains more facts and has greater predictability."

"We can only use our creative imagination to think up ideas and hypotheses," adds Hella, "and then we must quietly measure and experiment to see which verbal garment best clothes the world around us. It's a never-ending process."

As they lie on a responsive lounge discussing this aspect of scientific methodology, they are brought back to the here-and-now by the ten-minute landing signal. Now they begin to watch the water below. The coast of Florida with its silver lining of beaches recedes. The vibrant colors of the Bahama reefs appear, and, almost too soon, they arrive at their Exuma resort.

The blue-green panorama is interrupted as their craft lands on the top of the Exuma City in the Sea. This building is a spectacular engineering achievement. A large ring, or circular dam, rises from the bottom of the sea, which is fifty feet deep at that point. The structure projects 100 feet above the surface. The top of the circular city automatically slides open in dry weather to uncover cybernetically maintained recreation areas and tropical gardens with dining facilities offering a variety of delicacies from the Exuma Sound.

Scott and Hella select a room that is twenty-five feet below the surface. Large windows look out on the colorful underwater reef. When they enter their room, they instruct the cybernator for that unit to pick up the living patterns they have developed over the years with other cybernators in every place they have ever lived. Their preference for humidity, heat, light, music, food, and teleprojection program material are instantly available in their new home.

No Price Tags

Nothing in the twenty-first-century world of Scott and Hella has a price tag. Prices were a distribution mechanism that was inevitable in the scarcity cultures of previous centuries. The cybernated production and distribution complex of the twenty-first century is capable of producing many times the flow of goods and services that people of the world require. The automated production capacity is so great that if everyone on the earth were suddenly to order a portable teleprojection set, such an unlikely demand could easily be met with only a short delay.

In certain areas, however, there is a small "cost" to be paid, although neither Scott nor Hella thinks of it in these antiquated twentieth-century terms. They know that this underwater complex with 4,000 apartments requires a continuous staff of three people to operate it. Since there is not a single paid employee in the entire twenty-first-century world, they know in advance that they may be expected to contribute an hour of their time for each month they stay. They look forward to contributing this service, for it furnishes them with new experiences. All jobs involving drudgery have long since been cybernated so they know they will not be asked to scrub floors or perform boring menial tasks. They will probably stand by to help in any way they are needed. Whatever they may be required to do, they know it will probably be interesting, if not challenging.

Soon after arriving, they attend a one-hour teleprojection that gives information on the underwater complex. It shows some of the more popular types of activities; it outlines dangers and suggests certain precautions; it tells where and how to use underwater breathing apparatus and where to pick up their sub-mobile. It locates the various underwater parks that are within a three-hour range of their submobile and shows how to use a special computer to communicate with the intelligent, trained dolphins and other animals in the sea. There are demonstrations of underwater photography and the use of ultrasonically propelled water skis.

The teleprojection describes the magnetic field set up in the water on the north side of the building. Fish line up and swim toward the positive and negative poles of this electrical field. Pulses of high voltage herd them in groups toward a large funnel that sucks them into the cybernated processing plant. Aquatic plants are also grown in underwater fields, and the tops are harvested automatically, leaving the roots and lower third of the plant to grow a new crop without replanting. In various places throughout the world, local traditions often supplement the 325 varieties of food regarded as standard.

When the teleprojection is over, Scott and Hella pick up their handbook and board the underwater sightseeing craft that takes them on a ninety-six-mile tour of this colorful reef area. They frequently leave the submobile and use their membrane masks to explore underwater caves and grottos.

The Human Use of Time

That evening Scott and Hella join several men and women who are discussing some of the problems of

the previous century. No introduction is ever needed in the new world. Everyone feels outgoing and friendly toward his fellow man. The need for introduction in previous centuries often served as a status shield that maintained distance between people.

Myra, a petite blonde, is standing with her back to a large submarine window. She is the center of attention as she discusses the concern of their ancestors over the problem of what people would do with their lives when they didn't have to work. With vivacious movements she describes the dour predictions of the "emptiness of too much leisure." In a civilization of scarcity, it was customary for people to expect a life of unremitting toil and to develop "wisdom" based on this reality.

"If all the year were playing holidays, to sport would be as tedious as to work," Scott says with a smile, quoting from Shakespeare.

Anna, who is drying her hair with an air jet, remarks, "Historian Thomas Carlyle warned that 'a life of ease is not good for any man, nor for any god.' The folklore of the past was full of such admonitions as, 'Idle hands are the Devil's tools and idle minds, his workshop.' Our ancestors professed to have faith in humanity, yet they didn't trust people to direct their own lives."

"How incredible," says Daryl, "that humans could be so conditioned that they would feel guilty if they were not engaged in repetitive toil." He ambles over to the window to join Myra. "Why should people ever feel guilty about anything?"

"Somewhere around the mid-twentieth century," says Hella, "I recall that the Center for the Study of Democratic Institutions was studying the problem of what people would do with their spare time when they were no longer enslaved by the need to work for a weekly pay check. They invited Daniel Nugent, who lived on a hilltop nearby, to one of their weekly conferences. Nugent had owned a large department store in St. Louis, and he sold it out in 1916 when he was only twenty-seven. He retired in Santa Barbara and spent his days reading, studying, thinking, enjoying the loveliness around him, and using his money to help people. One by one the staff members in the conference room discussed the problems of what to do with leisure time in a world without work. 'What will happen when men's and women's lives are not structured for them? Can they make their own decisions? Can they use their own resources to build a worthwhile life?' Nugent sat there listening for a long while before he strongly protested, 'Gentlemen, I myself have not been gainfully employed for some 45 years—and I assure you there are not enough hours in the day.' "

"Nugent was a smart man," observes one of the older men in the group. "He put his finger on our real problem. Our lives are just not long enough in spite of the reduced time now needed for sleeping. It is impossible for any individual to experience even one-thousandth of the world that we have today. And all of our horizons are constantly expanding so that as civilization goes on, it seems that the individual can experience less and less of it."

"I agree," Hella responds. "Thomas Edison said, 'The stomach is the only part of a man which can be fully satisfied. The yearning of man's brain for new knowledge and experience . . . can never be completely met.' Perhaps if we could live 10,000,000 years we might find life boring. But Nugent was right—that certainly is not our problem today!"

Everyone laughs at this last remark. Amazing how people can get worked up over problems that don't even exist!

No Burdensome Possessions

After three weeks Scott and Hella realize they cannot leave soon. So many beauties, unique feelings, thoughts. Perhaps six months, perhaps a year would be enough. How could one decide in advance? They notify Central Correlation that they plan to remain here indefinitely. All conference calls and other communications are to continue to come to the Exumas, for they have cancelled their plans to return. They ask the Correlation Center to make their apartment available to other people. This is no problem, for Scott and Hella left no personal possessions there. In fact, they have few "personal" possessions. Whatever they want to use is available in any environment on earth.

The entire concept of personal possessions belongs to the old scarcity societies. It isn't that Scott and Hella are forbidden to have them. They don't want them. They have no need for them. All of the things that people of previous societies used and which are still functional in the twenty-first century are structured into the environment. Suppose someone were to tell Scott, "Here is a pen. It belongs to you. You must take care of it and not let anyone take it away from you when you're not looking." Scott would give him a what's-going-on-here reaction.

Besides, he has relatively little need to make marks on pieces of paper, for he can talk into the cybernator, and his words will be automatically recorded or printed. The finger-sized computer embedded in Scott's brain has sensory inputs that permit drawing by means of thought. If he wants to keep a copy of such a drawing, he instructs the cybernator to make a copy or to store the image he has created with his thought patterns.

Anything Scott and Hella want can be rapidly produced to their personal specifications and usually delivered in several hours, no matter where they are—on the earth, below the earth, or in the satellites above the earth. They would regard it as an imposition if they had to regard certain things as their own—to keep track of them, to take them where they go so that they would have them available when needed, to make sure that they are properly serviced and in good working order. What a crude bother! In contrast, Scott and Hella have everything they need anywhere on earth. They are never concerned with taking care of any physical objects, for maintenance is cybernated. "The old concept of ownership sounds utterly barbaric," Scott once observed. "It's burdensome and boring."

Artistic Expression Is a Part of Living

Scott and Hella find themselves deeply moved by the color of the reefs and waters, the savage brutality of the more aggressive fish, and the graceful motion of the marine plants and animals. While Hella is visiting the observation deck high above the Exuma Sound, she has a desire to express her feelings in a three-dimensional painting. She tells the cybernator of her wish to paint and walks over to a three- by four-foot panel. She picks up a lightweight instrument about three times the size of a pen. Through controlling the

adjustment on this instrument, she is able to produce any color, or mixture of color, desired. Just as a sliding trombone can produce graduations of pitch, so her electronic brush can produce a thousand different hues and tints. Fine lines are drawn by holding it close to the screen. Broad lines are made by pulling it back. The pen can paint a flat, two-dimensional picture, or it can build up the material into any three-dimensional pattern desired. If Hella is dissatisfied with her work and wishes to start over, she has only to indicate so, and the cybernator will electronically erase it. When Hella is through with her painting, she instructs the cybernator to record it. If she especially likes it, she will order the cybernator to transmit it to the Correlation Center.

Scott has a particular talent in sculpture, and he is inspired by the living forms surrounding him in the Bahamian waters. By using an electroformer, he is able to produce sculpture that in previous centuries might have involved days of hacking away at wood or stone. When he is satisfied with one of his productions, he orders the cybernator to send it to the Correlation Center. The physical structure of the sculpture is not moved, but through electronic scanning its contours and colors are recorded and transmitted. The Correlation Center schedules the exhibition of paintings and sculptures. Through three-dimensional teleprojection Scott's sculpture will probably appear for ten-minute intervals in several apartments, walkway areas, and research laboratories during the next week. The degree to which it will appear again in other areas of the world and whether it will ever be shown at a Cultural Center depends on the amount of attention it receives as recorded automatically by attention scanners.

If Scott or Hella were curious regarding the fate of their creations, they could ask their cybernator to request that information from the Correlation Center. They do not, however, produce these creations for the ego satisfaction of exhibiting them to others. They make them for their own pleasure. They create them because they have an inside need to express themselves. They produce them for the satisfaction of developing their artistic talents to a higher level. Whether anyone else in the world likes or dislikes their art is of little concern. Their main reason for transmitting their better products to the Correlation Center is to share with others something they feel would add slightly to the lives of fellow humans.

The Achievement of Liberty

That evening Scott and Hella join a group attracted by the panorama seen through a transparent wall fifty feet under water. They are immersed in a living symphony of fish and plant life. As they soak up the details of this brilliantly lit section of the reef, they have deeper insights into their cultural heritage. They watch the wanderings of a small shrimp as it scans its environment for food. Suddenly a snapper darts past, opens its jaws—and whack! The shrimp is no more. One's attention is arrested by the graceful co-ordination of the eight arms of a small squid. Suddenly a jack comes along and grabs it in the middle. The arms flail helplessly around the mouth of the jack. Then the jack is attacked by a barracuda, and the squid is immediately dropped as the jack flees for its life. The barracuda reaches down and grabs the now-injured squid with its sharp teeth. In three shakes it is devoured. Scott and Hella are impressed by the ferocity of life in the marine jungle—the cruel workings of the survival of the fittest, the inevitable conflict that is brought on by scarcity.

"Blessed are the meek," Scott quotes. "But the meek may not survive in the jungle. If animals or people

have to fight each other to get what they need, they become brutal. They have to be callous and heartless—it would tear them apart if they empathized with the pain of others."

"How indebted we are to our ancestors for working through those primitive stages so we can at last live as human beings," remarks Hella as she watches a playful sergeant major darting around a lavender-tinted lettuce coral. "They had the illusion of freedom—we have real freedom."

"Only recently have we been really free of the age-old ruts and routines," Scott continues. "Free from economic struggle, from aggressiveness in a million forms, from constant ego attack, and from always being told what to do. Even when our ancestors had enough food in their stomachs and had a roof over their heads, there was still a scarcity of love, affection, and emotional security to meet their ego needs."

"Yes," adds Hella, "and previous societies had intricate ways of giving status to people that enabled them to one-up the other fellow—to try to get a feeling of worth by showing that they were better in some ways than other fellow humans."

"I suppose most of the problem revolved around scarcity," says Scott. "People must feel secure to give deeply to others."

"They tried to get security by passing laws," Hella smiles. "I understand that in previous centuries thousands of laws were passed each year telling people what they could and could not do."

"It's been years since taboos or laws were forced by society on the individual," Scott says. "Previous cultures used to label various things as right or wrong, good or bad, moral or immoral, lawful or unlawful. These things sometimes changed from one state to another, from one country to another, and certainly from one culture to another."

"We shouldn't be too proud," admonishes Hella. "It was only two decades ago that we were able to dispense with the last law, the last lawyer, and the last courtroom. Only in our age could we be sure that human beings could be fully trusted if they are reared in ways that avoid hostile conditioning. Happy, fulfilled people never commit crimes!"

"I'm not sure that it's all a matter of trusting people," counters Scott. "I'm not sure I could be trusted not to harm myself or someone else if I were put into one of the automobiles of the last century. We've used technology to avoid hurting ourselves or others. Try to imagine, Hella. They had no automatic controls. They sped along those narrow highways. The death rate was horrible, the injuries even worse. In the United States auto accidents killed more people each year than their wars! This slaughter was so unnecessary. It's been decades since one of our surface transportation units injured anyone."

"The availability of the medium-range aircraft that we used to come here was shelved for four years until proximity control devices were perfected," says Hella. "This safety system reduces the probability of a crash to less than one chance in six trillion miles. The danger of a crash is more remote than being hit by lightning."

"Yes, I remember reading about the probabilities of a crash on the nameplate as we entered the craft," answers Scott. "There's no 'Big Brother' making decisions for us. We're given the facts and probabilities,

and we make our own choices."

"Watching those fish out there," says Hella, "makes me realize how far man has come. We can truly be ourselves—think what we want, feel what we want, experience what we want—without hurting other people."

Trip to an Underwater Park

The next morning Scott and Hella take one of the submobiles and navigate to an underwater park about twenty miles away. On the way they see the forms of many wrecked ships, now deeply encrusted with coral. They play tag with a dolphin for a while. In their large bubble enclosure in the forward part of the sub-mobile, they have full visibility. Built-in televised binoculars with microviewers enables them to examine marine life in detail. They call the Correlation Center via relay satellite and request a summary of the scientific work now being done in oceanography. A briefing keyed to their intermediate level of understanding is given to them as they near the underwater park.

In the park they find other submobiles. Scott and Hella having their underwater breathing membrane on, slip their feet into self-propelled fins, adjust their voice communicators, and exit through the air lock. For hours they explore the marine gardens and make three dimensional teleprojections with their laser cameras. They stay together so that they can help each other in case of emergency. Trained porpoises are there to assist if needed.

On the way back from the park, Scott and Hella direct their submobile to navigate automatically. He helps her remove her fins and breathing membrane. Caressing her ear with his lips, he peels her underwater suit from her body. With a knowing smile she turns and sees that he has already removed his gear. A word to the cybernator brings forth sensuous music with strong repetitive rhythms. She shares his mounting excitement.

Competing with Oneself

Scott and Hella still feel exuberantly energetic even after the underwater trip. They decide to play a game that was adapted from ping-pong of the previous century. The net, the table, the ball, and the paddles are almost unchanged. Their adversary, however, is very different, for Scott and Hella play on the same side of the table as partners against a mechanical paddle directed by a computer. This computer has sensing devices which enable it to judge the direction and speed of every ball returned over the net. Although the computer is able to return every ball with 100 per cent accuracy, it does not do so. The Correlation Center has a record of every time Scott and Hella have played this game as a team, and it has established a norm for them. At this time the norm for the preceding year indicates that Scott and Hella returned 85.967 per cent of the ping-pong balls that were directed to them with an average speed of 7.72, as measured on a 10 point scale. The computer plays a game against Scott and Hella that represents a skill exactly equal to the average of all the games that they have played during the preceding year. If Scott would want to play alone, a different set of records maintained by the Correlation Center would enable the computer to put up a game against him that would be exactly equal to his average performance.

Scott and Hella are thus able to play as a team—against themselves as a team. If they are in good shape

today, they will win. If they are not, they will lose. Either way they win, for no matter how the game comes out, they feel good. They have a lot of fun laughing and trying to figure out how to get around the computer.

They enjoy competing against themselves. They would find it repulsive to compete against each other. Such a battle would prove nothing. It could only be damaging in some slight way. If they compete against their past performance, they can tell if they are improving.

During the evening Scott reclines in the massaging contour chair in his underwater apartment. The "bay window" is illuminated so that occasionally he looks out to see if anything is happening on the reef. As he lies relaxed in his chair, he watches a screen that is angled above him where the paragraphs of a book flash by. Scott's usual reading pace is 22,000 words per minute, but he has slowed down to 7,000 words because he enjoys the languorous intermixing of the abstract thoughts of the book with the colorful underwater world beyond his window. Suddenly, he has an interesting idea: could a range emitter be designed that would repel sharks, barracuda, moray eels, and other marine animals that could endanger swimming? He wonders if this could be built into a lightweight belt for underwater use. He immediately calls Central Correlation and gives his thoughts in detail. Central Correlation then sends this information to men and women who are interested in this area. They will probably have a teleprojection conference sometime in the next week or two to discuss it.

The Disarmament Anniversary

That evening there is a world-wide ceremony scheduled by the Correlation Center. This marks the eighty-second anniversary of the date when the last instrument of death was destroyed. Previous cultures developed a long progression of tools designed to kill fellow human beings. It started with the cave man and his club. It ended with an ultimate weapon that could wipe out all life in an instant.

Scott at one time visited a museum and was appalled that human science and ingenuity could have been applied in such self-destructive ways. He was amazed that humans could have been that hostile toward each other. But he realized that he should not judge other people and other civilizations, for they had problems of which he was only dimly aware. He knew that had he lived in previous times, he probably would have played a part in piloting an airplane to drop a bomb or in rushing up a hill with a gun in his hand to kill the defenders at the top.

No one today wants instruments for killing. In the areas where wildlife exists, man finds no need to kill animals. He protects himself by using computers that communicate with animals in ways that control their behavior.

How remarkable human beings are, after all, Scott thinks. People in the past could live in a threatening world subject to being killed at any minute at the whim of a dictator in a foreign country. And yet they still managed to make a life of it and come through it to develop the present civilization. Scott wonders whether his nerves could have withstood this type of pressure. Would he, too, have developed the neurotic personality, the deep insecurities, hostile aggressiveness, hollowness of ego, and the scramble for a feeling of worth that characterized his ancestors? He is sure he would have in previous times. Fortunately, these are only words to him now. It is even difficult for him to be sure that he is using them in

a way that represents the feelings of people who used the same words in previous centuries.

Our Only Enemy

That evening Scott and Hella find a group in front of the large submarine window at the thirty-foot level. They have reason to ponder the long road ahead of them. The Correlation Center recently released their figures showing the degree to which humans are currently developing their intellectual potentials. It pointed out that during the previous century people in the more advanced civilizations used from 2 to 5 per cent of their mental capacity.. Recent measures showed that the people of Scott and Hella's world are using 18 per cent of their intellectual, artistic, and sensory capacity. No one knows at this time whether a higher percentage would result in more or less happiness. Further research is needed.

One of the older men in the group remarks that it is comforting to know that all things in their civilization are related to the needs and feelings of individual men and women. The method of science is used to measure the reactions of people. It is never used to force individuals to conform to any given, predetermined goals. Every program of improvement is carefully tested before it is adopted. Even after adoption it is still on probation forever. For nothing would be permitted to remain in succeeding centuries that would not contribute maximally to the happiness of the individuals who will be alive then. Man is the measure of all things, and the scientific method is the measurer.

"At last," comments Hella, "we have a civilization where the entire range of human needs can be met. For the first time in man's long history we can have complete diversity. It's remarkable what happens when you let people be themselves and do what they want."

"We certainly are fortunate," says Scott, "that we can live in so many dimensions. Our ancestors felt proud if they were experts in one or two things. My great-grandfather was a nuclear physicist who developed mathematical games as a hobby. He was considered quite brilliant because of his achievements in these two areas. The average person today enjoys operating in over 100 different areas."

"So many new fields are opening up. There is just not enough time."

"Time is our only real enemy," replies Scott with a frown. "Perhaps we'll lick it some day."

10. Designing the New Generation

After seven months in the Exumas, Scott receives a message from the Correlation Center that there is an opportunity at a medical laboratory in Calcutta, India. This laboratory specializes in designing and training the new generation—probably the most important function of the new society. Hella shares Scott's excitement at this chance to participate.

The query from the Correlation Center—affectionately called Corcen—is completely optional. Corcen never tells people they have to do anything. It simply presents information on opportunities that are available or situations that need attention. Each individual makes his own decision regarding what he wants to do. One might draw a rough analogy with an invitation to join an athletic team of the previous century. An invitation to be on the football team was regarded as an opportunity. The coaches didn't force players to join the team.

"In ancient Greece the Athenians were fond of saying that although other states might know how to make better products, only Athens knew how to make human beings," Hella says seriously. "All any social system really makes, well or ill, is human beings. It will stand or fall on this."

Hella prefers to remain in the Exumas longer, and she decides not to accompany Scott to India. By three dimensional color teleprojection, they can still "be together" as often and as long as they want. Since there is plenty of time, Scott asks his cybernator to arrange for a thirty-foot sailboat for cruising to Miami. He suggests that several companions would be desirable. Within two days Scott, another man, and two women leave the Exuma Islands in the sailboat. This boat has been designed to be non-cybernated. Scott and his companions have the novel feeling of being in an environment in which the routine flow of activities is not automatically structured. They find it quaint to open cans to get food, to navigate with a chart and compass, to fish with a hook and line, and even to pump the old-fashioned heads in the boat. A few days of this rather primitive living are delightful. It reminds Scott of what some people said of New York in the last century—"It's a nice place to visit, but I wouldn't want to live there," These experiences give them a broader understanding of how the lives of their ancestors were consumed so largely by the mechanics of living that they often had little time for intellectual, aesthetic, or sensual development.

"I suppose if I spent most of my waking life coping with these mechanics of living," Scott confides to one of the women, "I'd probably be too busy for much else."

The only part of this sailboat that is different from its counterparts of a century ago is an automatic communicator that has been built into the forepeak of the boat. Scott and his companions are only dimly aware of its presence. It sends out a radio signal every ten seconds. This is picked up by an orbiting satellite and relayed to Corcen. No human keeps a record of their location— only Corcen. If the signal from the boat were ever to stop, an immediate attempt would be made to contact Scott through a built-in alarm. If this were to fail, some craft flying over the area with vertical capacity would be alerted by Corcen to give immediate assistance to the sailboat. All of this could be automatically programmed by Corcen. In a simple rescue situation, the only humans that would know about it would be those aboard the rescuing craft and those being rescued.

Lazily pushed by the prevailing southeast breezes, Scott's sail-boat makes its way over the White Bank south of Nassau. As they arrive at the Tongue of the Ocean where the depth goes from about 15 feet to over 5,000 feet within a short distance, Scott thinks of requesting Corcen to furnish him with information on the research that has been conducted there. Then he remembers that there is no teleprojection screen aboard his small craft. In a way he is glad. It is nice to rely only on his own senses and his own experiences, to see the deep blue water, to observe keenly the panorama of shifting clouds against the pale blue sky, to see without words, to feel in silence. They anchor several days north of Andros Island to absorb this more fully. Swimming, fishing, nude sun bathing—their days have a different fullness. They have no contact with the outside world, and there are few other boats or humans to be seen.

After a few days they haul the anchor and head west across the Great Bahama Bank. This large underwater desert is seldom more than ten feet deep. The water is crystal clear. Although there are few fish, the surprise and delight of watching the countless starfish add additional threads to their rich tapestry of experience. Soon they pass the jagged rocks of Gun Key and head west across the Gulf Stream. Their southeast breeze holds firm, and ten hours later, tanned and exhilarated, they arrive in Miami.

On his way to the South Florida terminal, Scott notices someone -waving for help in a park near the road. He instructs the car to pull over to the side. He notifies Corcen of his location and that he is going to the rescue. The person waving for assistance takes him to a man whose leg has been crushed by a large fallen branch. They are unable to budge it. Scott runs back to his car and tells Corcen. Corcen immediately gives this emergency information to several people in the vicinity. Within minutes six people lift the branch and release the injured man. They carefully carry him to a car which speeds him to the nearest cybernated hospital.

This instant helpfulness is an important ingredient in the new society. In former times people often felt, it's someone else's job, I don't even know him, he might sue me, why get involved." Such reactions may have been appropriate in competitive, legalistic, money-motivated societies. In the twenty-first century people welcome the opportunity to assist others. Is there a more human way to spend time? Everyone is more secure when he feels that everyone else in the world genuinely welcomes the opportunity to be of assistance. Normally, people need so little help in this cybernated world of abundance that this open-hearted feeling towards others never results in one's being overburdened. This generous willingness to pitch in and help extends beyond simple emergency situations. Each person identifies with the whole of society. If someone notices that some equipment needs repairs, he does what he can to fix it or reports it to Corcen. People treat all objects with the care and consideration that individuals previously gave only their own possessions. These individuals who have "everything" are able to give generously of themselves. Only in this century have all men and women so fully identified themselves with everyone and everything in the world.

The Great Circle Express

The simplest way to get to India is to board one of the around-the-world express crafts that has been continuously circling the globe without landing for many decades. They travel at a steady speed of 5000 miles per hour. When this craft is approximately one thousand miles away, Scott takes off in a shuttle craft that accelerates to 5,000 miles per hour. When the nuclear-powered, circumnavigating craft is above South Florida, Scott's airship locks onto the larger craft just long enough for Scott and other passengers to get aboard. Departing passengers enter the shuttle craft, which then disengages and returns to the South Florida terminal a few minutes later.

The flight to India takes a little over two hours. From his vantage point high in the sky, Scott enjoys the planned geometry of the world below—the vast waterways that have eliminated forever the tragedy of floods, the cybernated farm belts with their thin gleaming tracks, the dynamic cities that are focal points of a technology harnessed to serve all of mankind. As Scott watches the cities of India, he is impressed by

how often the circular plan has been used. He sees a multi-story apartment ring about a mile in diameter under construction. Cybernated building machinery is fabricating 5,000 apartments a day with a construction staff of only thirty-five people.

The cities of the previous century have been leveled except for several that have been kept for anthropologists and historians. These museum cities are protected by a large, transparent, geodesic dome and are air conditioned. Similar protection is given selected ruins of previous cultures.

As Scott nears Calcutta, the shuttle craft ahead streaks upward for its rendezvous. It locks onto the larger space vehicle over Calcutta where a brief exchange of passengers takes place. Scott boards the smaller craft and within minutes alights at the Calcutta airport. There are no porters, for passengers have no baggage. There are no customs agents since international divisions have been meaningless for many decades. Scott confers with a local cybernator about apartment availabilities while he waits for his cybernated auto.

In Calcutta, Scott chooses an apartment with nine other men and women companions. He could have a private apartment but he is in the mood for group living. He gives instructions to the cybernator in the apartment to pick up his system of preferences from Corcen. Thus, within the limits of the preferences of his companions in the apartment, Scott will continue to feel perfectly at home.

Circular Cities

Many of the colorful cities in the twenty-first century are laid out in the circular pattern Scott observed from the air. The central hub of the city has a nuclear generator that produces all of the power needed to operate the millions of invisible electronic servants that silently free the people to live fulfilling, creative lives. The central core of each city also contains a master computer that cybernetically watches over the city as a whole. It is connected to every room in the entire city and also Corcen. This master cybernator, operating as the city hall of former times, works automatically and normally has no humans in attendance.

The research laboratories are located in the first ring of buildings that encircle the power and computer core. Hospital facilities adjoin the medical research area. These modern hospitals give medical care or nursing care to a population of 1,000,000 people with a staff of only 10, who contribute their services from time to time because of their interest in this work. There is, of course, much less illness in the new world than in previous centuries, and there are practically no accidents. Disasters and accidents were almost eliminated by an attitude toward engineering that minimized economy and maximized safety. All diagnostic techniques, laboratories, surgery, behavioral assistance, and nursing procedures are cybernated. The small staff performs no routine duties in the cybernated hospital. They are only there to help in the rare event of a problem.

The second ring from the core contains multi-story apartments. They are over a quarter-mile from the research ring. Recreational facilities and circular parks surround the apartments on both sides of the ring.

When Scott goes from his apartment to any part of the city, he asks the cybernator to summon an auto.

As Scott approaches the vehicle, the door automatically opens and the seat swings out. Scott verbally gives his destination and reclines on the contour chair. He is then free to read, to think, or simply to relax. Within minutes the auto chauffeurs him to wherever he wishes to go in the city. All vehicles have proximity mechanisms for the prevention of what were formerly called "accidents" but are now considered technical negligence in planning the transportation system.

These autos are available throughout the entire city for the use of everyone. When Scott gets out at the research laboratory, the auto is directed by the city's cybernator to the next point at which it will probably be needed.

Scott always enjoys the warm feelings of teamwork and cooperation that are experienced by people working together on common problems. There is an *esprit de corps*—a feeling of man against the unknown; a feeling of contributing significantly to the present and future happiness of all mankind. People in the twenty-first century are eager to accept opportunities for participation and research. They are never paid for this activity as had been common in previous centuries. How could they be paid? They already have every material resource of the twenty-first-century civilization available for their use. They are not even "paid" by prestige or status.

Almost everyone at one time or another plays a part on various research teams. If someone doesn't, it really doesn't matter. The only reward lies not on the outside, but on the inside. It comes from the pleasure that one gets from exercising his mind, from growth and improvement, from the pleasures of understanding, and, from the contented feeling of saying, "Well, we certainly solved that one."

The Genetics Laboratory

The laboratory to which Scott is assigned specializes in the manipulation of the DNA and RNA structures of human genes. The people of the twenty-first century have worked out techniques for varying the structure of the human body. By using a computer to change various sets among the five billion specifications carried by DNA and RNA molecules, almost any change may be made in a human body. Their major concern at this point lies not, for example, in how to equip a human being with two hearts instead of the usual one, but in whether such an arrangement will add to human happiness.

"All changes made in the structure and function of human beings are first thoroughly tested in research labs," Scott is informed. "Then experimental and control groups are set up to get valid comparisons. No improvements are considered desirable in their own right. No guessing is permitted. It has been found that something that can seem like a great idea might turn out in practice to be nothing of the sort. All ideas for the genetic improvement of human beings are thoroughly tried out, usually over a period of decades, before they are generally adopted for programming into the new generation. The door is always left open. 'Blueprints' for genes are always stored in case future generations evaluate differently and wish to eliminate any changes made."

Scott is quickly brought up to date on research in progress. Trial runs involving 500 people are now being made with brain structures containing twenty billion neurons—double the usual number. Other projects in progress are designing eyes that can shift from regular vision to telescopic and microscopic vision and

altering the liver to change the composition of the blood in a way that seems to add 36 per cent to the human life span.

They are experimenting with an improved hormone balance for women that eliminates the monthly cycle of moodiness— menstruation was eliminated previously by designing a uterus with a stable lining. Men have also benefited from the improved techniques of genetic manipulation. A greater climax frequency now enables them to perform at the high level usually desired by twenty-first century women.

Scott knows that in previous times there were five races of mankind. Since individuality and diversity are prized in the twenty-first century, these genetics laboratories have produced eight more races. Corcen is seeking data to determine whether additional races should be designed to add even more variety to the lives of future men and women.

Suppose a part of the body should wear out or get injured. How could we get a duplicate organ? Each cell in the body contains a blueprint for growing a replacement part. Research is in progress to use a cell from the body of an injured person to grow an identical part in-vitro that a cybernated surgical mechanism could install.

One of the most exciting new developments is a built-in receptor that makes it possible to connect one's brain directly with Corcen or any other input source. If invited, you can tune into another person's brain and share his thoughts and feelings without the distorting effects of words. When this is perfected, any sensations can be experienced through direct neuronal input.

Other researchers are developing a lifetime implanted communication unit that would permit two-way thought messages. By thought, one could request from Corcen a bit of information, and it would be available immediately. Such thought communication with others would, when perfected, offer an electronic realization of what used to be called "mental telepathy."

Breakthroughs are being made in controlling the factors that permit cells to age. Aging is considered a disease by these men and women. They are confident that when it is fully understood, it can be eliminated. A "youth serum" that includes thyroxine and a mixture of hormones has doubled the years of vitality and has added an average of 89 per cent to the life span in a test on laboratory animals.

"Humility when facing the unknown is our dominant theme," one of the older men says to Scott. "We are producing things that by the standards of previous centuries would have been regarded as fantastic miracles. Yet, no matter how successfully our experiment turns out, we always leave the door open for improving the results. We never believe we have the best way. We always have the feeling, 'It seems to be working well now, but it hasn't stood the test of centuries or millenia. Let's go slowly and not close any doors behind us.' "

Brain Boosters

John F. Kennedy said in the past century, "The human mind is our fundamental resource. . . ." Scott knows that the greatest achievement of these genetic laboratories during the past half-century was the implantation of a finger-sized organic computer in the growing embryonic brain. Everyone in the new

society that is under fifty years old has this new development. Since Scott is only 45, he has the benefit of this breakthrough in designing human beings. By manipulation of the DNA and RNA molecules, a small auxiliary brain was developed that is nurtured in-vitro outside of the human body.

When the cortical cells of this supplementary brain complete their proliferation, they are electronically connected with Corcen. These brains are then imprinted with the basic attitudes and skills needed for orientation in the twenty-first century. Scott watches as the cybernated facilities program the small but potent brain boosters. To the three R's of previous times—reading, 'riting, and 'rithmetic—an additional seven R's, as outlined by Dr. William A. McCall in the twentieth century, are included:

RESEARCH
The application of the scientific method as a way of life that enables man to test ideas to determine their reliability.

REASONING
The desire and ability to manipulate ideas in a creative and logical way. Adjusting constructively to new situations and making effective choices.

RELATIONSHIPS
Attitudes and skills that enable a person to interact and communicate with others in ways that bring maximum mutual satisfactions.

REPORTING
The use of all senses in ways that produce the richest and most accurate input to the brain and output to others.

RECREATION
The attitudes and skills which permit the use of one's time to achieve a multi-dimensional life.

REVITALIZATION
Diet, health, and safety skills that add years to life, and life to years.

RESPONSIBILITY
The feeling of playing an important part on the human team in the game of life. The ability to find satisfaction in assisting and participating—but always within the limits one can give without resentment.

As suggested by McCall, each of these R's is two-phased—an attitude phase and a skill phase. It is not enough that the young be provided with the skills of reasoning. It is equally important that they find pleasure in reasoning. It is not enough that the young know how to read. They need to enjoy reading.

Scott often uses his supplemental brain as a simple computer.

He can multiply, divide, add. or subtract any six digit number in a period of ten seconds. A complete vocabulary and understanding of the grammatical structure of the universal language is also imprinted.

In addition to being provided with an array of the basic tools and skills needed for orientation, this supplementary brain is also imprinted with information equivalent to a Ph.D. level in twelve different areas of learning. These areas are selected at random by Corcen in such a way that few individuals have the same pattern of intellectual development. Corcen also selects one area of learning and imprints this supplementary brain with every bit of information in this area that has been accumulated in the extensive memory banks.

For instance, if a supplementary brain is chosen to receive "complete" information in the field of anthropology, it is imprinted with a word-by-word reproduction of every worthwhile article ever published in that field that has been recorded in the memory banks of Corcen. It is imprinted with every book, every recorded lecture by eminent people in the field of anthropology, extensive simulated field experience, plus a briefing of all work now in progress. This is an internal treasure of knowledge that the individual can never live long enough to exhaust completely. But it will always be there in his brain, available for use to the extent that the individual can use it.

After these supplementary brains are matured and fully imprinted with this enormous assortment of attitudes and skills and their subordinate informations, they are attached to a growing embryo at a time of rapid proliferation of the ectoderm. When the ectoderm begins this stage, the supplementary brain is rapidly absorbed and integrated into the human nervous system. Since this implantation cannot be performed satisfactorily in an embryo inside the body of a woman, babies are grown from DNA engineered germ plasm in the cybernated "uterine" containers.

The sperm and egg cell used by the human race in the long evolutionary past are no longer needed. Reproductive cells are produced in the laboratory that are designed to develop into vastly improved versions of *Homo sapiens.* These cells are engineered by Corcen and can be programmed to develop into male or female embryos. At the age of approximately nine months, the developed infant is removed to continue its growth in the cybernated nurseries.

A woman of the twenty-first century does not want an infant to emerge from her vagina any more than a man in previous times would have desired a baby to grow in his body. Just as no man or woman in the twentieth century would knowingly have brought into the world an imbecile baby, it would be equally abhorrent for any twenty-first-century man or woman to bring into the world a baby that is not equipped with this supplementary brain. Such a person, even though possessed with an I.Q. equal to Einstein's, would feel like a moron relative to his companions.

Scott is aware that the implantation of this brain booster does not automatically produce an infant who can solve problems in calculus. The resource is there—available but untapped—just as the capacity of young Mozart existed at birth even though his infant fingers had never touched the keys of a piano. Only maturation can provide the experience and motivation that enables these young infants of the twenty-first century to make use of their great heritage.

The Child Is His Best Teacher